Project Title

Quick Commerce Operations Analysis Using SQL

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1. Introduction

Quick commerce platforms aim to deliver products to customers within a very short time, typically between 10 and 30 minutes. To achieve this level of efficiency, companies must constantly analyze operational data such as orders, delivery times, customer activity, and product demand.

Companies such as Blinkit, Zepto, and Swiggy rely heavily on data analysis to optimize logistics, inventory management, and customer experience.

This project uses SQL to analyze quick commerce data and extract meaningful insights about customer behavior, product demand, delivery performance, and revenue generation.

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2. Objectives of the Project

The main objectives of this analysis are:

- Analyze the number of orders placed on the platform.
- Identify top-selling products.
- Measure average delivery time.
- Determine revenue generated by different product categories.
- Identify cities with the highest order volume.
- Analyze customer purchasing behavior.

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3. Tools Used

The following tools were used in this project:

- **MySQL Workbench** – Database management and query execution
- **SQL** – Data querying and analysis
- **CSV datasets** – Source data for the database tables

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4. Database Structure

The project database contains four tables.

Customers Table

Column	Description
customer_id	Unique identifier for customers

Column	Description
customer_name	Name of the customer
city	Customer location
signup_date	Date of registration

Products Table

Column	Description
product_id	Unique product ID
product_name	Name of the product
category	Product category
price	Product price

Orders Table

Column	Description
order_id	Unique order ID
customer_id	Customer placing the order
order_date	Date of order
delivery_time_minutes	Time taken to deliver

Order_Items Table

Column	Description
order_id	Order reference
product_id	Product purchased
quantity	Quantity ordered

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5. Data Analysis Queries

Several SQL queries were performed to analyze the dataset.

Total Customers

total_customers
50

```
SELECT COUNT(*) AS total_customers
FROM qcommerce_customers;
```

Total Orders

total_orders
100

```
SELECT COUNT(*) AS total_orders
FROM qcommerce_orders;
```

Average Delivery Time

avg_delivery_time
16.4600

```
SELECT AVG(delivery_time_minutes) AS avg_delivery_time
FROM qcommerce_orders;
```

Top Selling Products

product_name	total_quantity_sold
Apples	106
Tomatoes	99
Potatoes	97
Bread	86
Milk	85
Eggs	78

```
SELECT p.product_name,
SUM(oi.quantity) AS total_quantity_sold
FROM qcommerce_order_items oi
JOIN qcommerce_products p
ON oi.product_id = p.product_id
GROUP BY p.product_name
ORDER BY total_quantity_sold DESC;
```

Revenue by Product Category

category	total_revenue
Fruits	586.8000000000001
Vegetables	451.5999999999999
Dairy	446.5
Grains	363
Snacks	299.79999999999995
Bakery	154.79999999999998

```
SELECT p.category,
SUM(oi.quantity * p.price) AS total_revenue
FROM qcommerce_order_items oi
JOIN qcommerce_products p
ON oi.product_id = p.product_id
GROUP BY p.category
ORDER BY total_revenue DESC;
```

Orders by City

customer_name	total_orders
Customer_4	6
Customer_1	5
Customer_43	5
Customer_5	4
Customer_38	4
Customer_8	4

```
SELECT c.city,
COUNT(o.order_id) AS total_orders
FROM qcommerce_orders o
JOIN qcommerce_customers c
ON o.customer_id = c.customer_id
GROUP BY c.city
ORDER BY total_orders DESC;
```

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6. Key Insights

From the analysis, several insights were observed:

The platform processes a significant number of orders, indicating active customer engagement.

Some products are ordered far more frequently than others, suggesting strong demand for certain grocery and snack items.

Average delivery time remains within a short range, supporting the quick-commerce promise of fast delivery.

Certain cities generate a higher number of orders, indicating stronger market demand in those locations.

Specific product categories generate higher revenue, which can guide inventory and marketing strategies.

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7. Business Implications

The findings of this analysis can help quick commerce companies improve operational efficiency.

Understanding top-selling products can help optimize **inventory management**.

Analyzing order distribution across cities can help companies focus on **high-demand regions**.

Monitoring delivery times helps ensure companies maintain their **fast-delivery service promise**.

Revenue analysis by category can guide **pricing strategies and product promotions**.

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8. Conclusion

This project demonstrates how SQL can be used to analyze operational data in a quick commerce environment. By querying customer, product, and order data, valuable insights can be extracted that help improve logistics efficiency, product availability, and customer satisfaction.

SQL plays a critical role in enabling data-driven decisions for modern quick commerce platforms.

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9. Future Improvements

Future analysis could include:

- Delivery partner performance analysis
- Time-of-day order trends
- Customer retention analysis
- Predicting demand for products